

Agentic AI & RAG Fundamentals

1. Agentic AI Systems

Agentic AI systems use **Large Language Models (LLMs)** as reasoning engines to plan, orchestrate tools, and make autonomous decisions.

2. RAG Architecture

A **Retrieval-Augmented Generation (RAG)** system combines context retrieval with generation to produce grounded and accurate responses.

3. Vector Retrieval Component

The retrieval component searches a **Vector Database** to fetch relevant semantic chunks based on similarity matching.

4. Vector Store & Embeddings

ChromaDB is an open-source vector database used to store high-dimensional embeddings and metadata for fast similarity search.

5. Advanced Retrieval: HyDE

HyDE (Hypothetical Document Embeddings) improves retrieval by generating a hypothetical answer before searching the vector database.